

From 2+1D Nonlinear Klein–Gordon Energy Lumps to Geodesic Propagation in an Effective Metric Induced by Substrate Inhomogeneity, Anisotropy and Flow

A Construction-Site First Draft Generalizing the 1+1D Sine–Gordon Synthesis

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Technical Synthesis Notes

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Abstract

We initiate the systematic generalization of the 1+1D sine–Gordon synthesis of 14 August 2026 to two spatial dimensions. The continuum is now a 2+1D nonlinear Klein–Gordon field whose potential $U(u)$ is left generic (an even function of the field, expressible as a series of even powers) but assumed to admit localized energy-carrying solutions. The most general Lagrangian density already incorporates a macroscopic substrate-flow velocity field $\mathbf{V}(\mathbf{x}, t)$ through the material (convective) derivative $D_t u = \partial_t u + \mathbf{V} \cdot \nabla u$. The constitutive structure is enlarged: the torsional stiffness becomes a symmetric positive-definite rank-2 tensor $\Theta^{ij}(\mathbf{x}, t)$, while the inertial density $\rho(\mathbf{x}, t)$ is retained as a scalar. Under a natural impedance-matching condition that keeps a single scalar impedance Z_0 constant, a tensorial refractive-index field can be defined.

Exact closed-form solutions analogous to the sine–Gordon breathers are not available in general; existence of finite-energy localized solutions (hylomorphic solitary waves, Q-ball-like objects, or numerically observed breathers) is guaranteed under standard hypotheses by the theorems of Benci–Fortunato [2,3] and is supported by numerical experiments. For any such localized energy lump whose internal size is short compared with the gradients of the constitutive fields and of the flow (eikonal / adiabatic regime), a collective-coordinate reduction yields an effective relativistic particle Lagrangian living in a Lorentzian metric induced by the refractive-index tensor and the flow. The subsequent logical chain—einbein formulation, geodesic equation, gravitational time dilation, anisotropic length contraction, free-fall trajectories, and the gravito-inertial equivalence theorem—then proceeds exactly as in the 1+1D case. The present note records the necessary modifications to the Lagrangian density, the constitutive dictionary, the existence statement, and the eikonal hypothesis; detailed derivations of the induced metric components and of the collective-coordinate integrals are left for successive drafts.

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1 Motivation and Logical Chain

The baseline synthesis of 14 August 2026 [1] demonstrated, in one spatial dimension, how a classical non-relativistic continuum (the sine–Gordon substrate) produces

1. exact localized energy lumps (breathers) that behave as relativistic particles,
2. special-relativistic kinematics (length contraction, time dilation, de Broglie phase) as intrinsic properties of those solutions,
3. an effective Lorentzian metric induced by spatial inhomogeneity of the constitutive parameters and by macroscopic flow of the substrate,
4. geodesic motion of both massive and massless energy concentrations in that metric,
5. the equivalence principle recovered as a theorem of continuum mechanics [1].

The construction relied on three pillars that do not survive unchanged in higher dimensions:

- the exact integrability of the sine–Gordon equation and the consequent availability of closed-form breathers;
- the scalar character of both the inertial density ρ and the torsional stiffness Θ ;
- the purely one-dimensional geometry of the laboratory space.

The purpose of the present construction-site note is to remove these restrictions while preserving the overall logical architecture. We therefore replace the sine–Gordon continuum by a 2+1D nonlinear Klein–Gordon continuum whose potential is generic, promote the stiffness to a tensor, introduce a tensorial refractive index under impedance matching, and state the eikonal hypothesis under which the collective-coordinate reduction and the induced-metric argument continue to hold. Explicit analytic solutions are no longer required; existence theorems and numerical evidence suffice.

The resulting programme remains strictly bottom-up and ontogenetic: the substrate is a classical continuum living in a preferred Euclidean space and Newtonian time; special- and general-relativistic effects emerge as properties of localized solutions and of the effective geometry they feel.

2 Lagrangian Density in 2+1 Dimensions

Let the laboratory space be the Euclidean plane with Cartesian coordinates x^i ($i = 1, 2$) and let t be Newtonian time. The dynamical degree of freedom is a real scalar field $u(\mathbf{x}, t)$. The continuum itself may possess a macroscopic velocity field $\mathbf{V}(\mathbf{x}, t)$ relative to the laboratory (the local kinematic flow of the substrate). The most general Lagrangian density that incorporates this flow is written with the material (convective) derivative

$$D_t u := \partial_t u + V^k \partial_k u = \partial_t u + \mathbf{V} \cdot \nabla u. \quad (1)$$

The laboratory-frame Lagrangian density then reads

$$\mathcal{L} = \frac{\rho(\mathbf{x}, t)}{2} (D_t u)^2 - \frac{1}{2} \Theta^{ij}(\mathbf{x}, t) \partial_i u \partial_j u - U(u). \quad (2)$$

Expanded, the kinetic term contains the ordinary time derivative, the cross terms linear in $\mathbf{V}$, and the quadratic terms in the flow:

$$(D_t u)^2 = (\partial_t u)^2 + 2 V^k (\partial_t u) (\partial_k u) + V^i V^j (\partial_i u) (\partial_j u). \quad (3)$$

Here:

- $\rho(\mathbf{x}, t) > 0$ is a scalar inertial density (mass density of the continuum of oscillators, dimensions mass per unit area);

- $\Theta^{ij}(\mathbf{x}, t)$ is a symmetric positive-definite rank-2 tensor field that encodes the torsional (or elastic) stiffness of the continuum; its eigenvalues set the local wave speeds along the principal axes;
- $\mathbf{V}(\mathbf{x}, t)$ is the substrate flow velocity field (dimensions of velocity);
- $U(u)$ is a potential energy density that is an even function of u . We leave U implicit, requiring only that it admit a Taylor expansion in even powers,

$$U(u) = \sum_{k=1}^{\infty} \frac{a_{2k}}{(2k)!} u^{2k} \quad (\text{or an equivalent non-polynomial even function}), \quad (4)$$

with $U(0) = 0$, $U''(0) > 0$, and growth conditions sufficient for the existence of finite-energy localized solutions (see Sec. 4).

The corresponding action is

$$S[u] = \int dt \int_{\mathbb{R}^2} d^2x \mathcal{L}. \quad (5)$$

When $\mathbf{V} \equiv 0$ the Lagrangian reduces to the non-flowing form of earlier drafts. Variation with respect to u (treating the constitutive fields ρ , Θ^{ij} and the flow $\mathbf{V}$ as prescribed backgrounds) produces the Euler–Lagrange equation

$$\partial_t(\rho D_t u) + \partial_i(\rho (D_t u) V^i - \Theta^{ij} \partial_j u) + U'(u) = 0. \quad (6)$$

(The first two terms arise as the formal adjoint of the material derivative applied to the density-weighted material velocity $\rho D_t u$.) Expanding the products with the product rule yields the fully explicit form

$$\begin{aligned} &(\partial_t \rho) (D_t u) + \rho \partial_t (D_t u) + (\partial_i V^i) \rho (D_t u) + V^i (\partial_i \rho) (D_t u) + \rho V^i \partial_i (D_t u) \\ &- (\partial_i \Theta^{ij}) \partial_j u - \Theta^{ij} \partial_i \partial_j u + U'(u) = 0. \end{aligned} \quad (7)$$

In a homogeneous isotropic non-flowing medium ($\rho = \rho_0$, $\Theta^{ij} = \Theta_0 \delta^{ij}$, $\mathbf{V} = \mathbf{0}$) this reduces to the standard nonlinear Klein–Gordon equation

$$\frac{1}{c_0^2} \partial_t^2 u - \nabla^2 u + \frac{1}{\Theta_0} U'(u) = 0, \quad (8)$$

where the reference wave speed is

$$c_0 = \sqrt{\frac{\Theta_0}{\rho_0}}. \quad (9)$$

When the backgrounds are constant and $\mathbf{V}$ is a non-zero constant vector the linearised operator becomes the convective wave operator $(\partial_t + \mathbf{V} \cdot \nabla)^2 - c_0^2 \nabla^2$, whose characteristics define the acoustic metric of a uniformly flowing medium [4,5,6].

Remark on the density. In the continuum limit of a lattice of oscillators the linear density of oscillators along a cut can in principle be direction-dependent. One could therefore promote ρ itself to a tensor. Standard continuum mechanics and crystal acoustics [7], however, keep the mass density scalar and place all anisotropy in the stiffness tensor. We adopt the latter convention for the present draft; the tensorial- ρ alternative remains available if microscopic lattice arguments demand it.

3 Impedance Matching and the Tensorial Refractive Index

In the 1+1D theory the impedance-matching condition

$$\rho(x, t) \Theta(x, t) = \rho_0 \Theta_0 = Z_0^2 = \text{const} \quad (10)$$

eliminates reflection while still permitting refraction. The natural generalisation to d spatial dimensions that recovers the isotropic case is

$$\rho (\det \Theta)^{1/d} = \rho_0 \Theta_0 = Z_0^2 = \text{const.} \quad (11)$$

(The power $1/d$ is required because, for an isotropic tensor $\Theta^{ij} = \Theta \delta^{ij}$, one has $\det \Theta = \Theta^d$, so $(\det \Theta)^{1/d} = \Theta$.) For the present 2 + 1-dimensional theory one has $d = 2$, and the matching condition therefore takes the explicit form

$$\rho(\mathbf{x}, t) (\det \Theta^{ij}(\mathbf{x}, t))^{1/2} = \rho_0 \Theta_0 = Z_0^2 = \text{const.} \quad (12)$$

Here ρ_0 and Θ_0 are the constant reference values of the homogeneous isotropic medium, while $\rho(\mathbf{x}, t)$ and $\Theta^{ij}(\mathbf{x}, t)$ are the local constitutive fields.

In the homogeneous isotropic case $\Theta^{ij} = \Theta_0 \delta^{ij}$ the matrix is $\text{diag}(\Theta_0, \Theta_0)$, therefore

$$\det(\Theta^{ij}) = \Theta_0 \cdot \Theta_0 = \Theta_0^2 \quad \Rightarrow \quad (\det \Theta^{ij})^{1/2} = \Theta_0. \quad (13)$$

The matching condition then reduces to $\rho \Theta_0 = \rho_0 \Theta_0$, i.e. $\rho = \rho_0$, as required. (The power $1/2$ is essential: without it the left-hand side would have the wrong dimensions.)

Dimensional check: in two spatial dimensions $\det(\Theta^{ij})$ has the dimensions of (stiffness)², therefore $(\det \Theta)^{1/2}$ has the dimensions of stiffness (the geometric mean of the two principal stiffnesses). Multiplication by the areal density ρ yields the same dimensions as $\rho_0 \Theta_0$ (impedance squared). Both sides of (12) therefore have the MKS units of impedance squared and the equality is dimensionally homogeneous.

Under this constraint a tensorial refractive-index field $n_{ij}(\mathbf{x}, t)$ is defined explicitly as follows. First form the local **acoustic tensor**

$$A^{ij}(\mathbf{x}, t) := \frac{\Theta^{ij}(\mathbf{x}, t)}{\rho(\mathbf{x}, t)}. \quad (14)$$

(The acoustic tensor A^{ij} is the direct analogue, for a scalar continuum, of the Christoffel / acoustic tensor of crystal acoustics [7]. It is a true spatial tensor: its eigenvalues λ_a , measured in a local orthonormal frame, are the squared principal phase speeds. **Notation convention:** throughout this note the symbol $\det A$ always denotes the product of these *physical* eigenvalues. In Cartesian coordinates the distinction between the determinant of the matrix of coordinate components and the physical determinant disappears; in orthogonal curvilinear coordinates the scale factors of the coordinate basis must be restored before the determinant is evaluated—e.g. in polar coordinates $(\det A)_{\text{phys}} = A^{rr} \cdot (r^2 A^{\varphi\varphi})$.) From the definition of A together with the impedance-matching condition (12) one recovers the constitutive fields themselves:

$$\rho = \rho_0 \frac{c_0}{(\det A)^{1/4}}, \quad \Theta^{ij} = A^{ij} \frac{\Theta_0}{c_0 (\det A)^{1/4}}. \quad (15)$$

(The isotropic homogeneous limit is recovered correctly: when $A^{ij} = c_0^2 \delta^{ij}$ one has $\rho = \rho_0$ and $\Theta^{ij} = \Theta_0 \delta^{ij}$. The identity $c_0^2 = \Theta_0 / \rho_0$ has been used to eliminate Θ_0 from the expression for ρ and ρ_0 from the expression for Θ^{ij} .) The numbers λ_a ($a = 1, 2$) are the physical eigenvalues of A , i.e. the roots of the characteristic equation

$$\det(A^{ij} - \lambda \delta^{ij}) = 0, \quad (16)$$

or equivalently the solutions of the generalised eigenvalue problem

$$\Theta^{ij} e_j = \lambda \rho e^i. \quad (17)$$

Let $\mathbf{e}_a$ be the corresponding orthonormal eigenvectors; they determine the principal axes. The principal refractive indices are then

$$n_a = \frac{c_0}{\sqrt{\lambda_a}} = \frac{c_0}{c_a}, \quad (18)$$

where $c_a = \sqrt{\lambda_a}$ are the principal phase speeds and $c_0 = \sqrt{\Theta_0/\rho_0}$ is the reference wave speed of the homogeneous isotropic medium. Both c_0 and $\sqrt{\lambda_a}$ have dimensions of velocity, so n_a is dimensionless. The refractive-index tensor n_{ij} is the unique symmetric positive-definite tensor that possesses the same principal axes $\mathbf{e}_a$ and the eigenvalues n_a . By construction:

- the principal phase speeds are exactly $c_a = c_0/n_a$;
- in the isotropic limit $n_{ij} \rightarrow n\delta_{ij}$ one recovers the original scalar refractive index and the relations $\rho = \rho_0 n$, $\Theta = \Theta_0/n$ of the 1+1D theory;
- the matching condition (12) is maintained and a single scalar impedance Z_0 remains well-defined.

The inverse map (expressing ρ and Θ^{ij} in terms of n_{ij}) follows at once from the spectral definition together with the matching condition.

Refractive-index tensor as a function of the acoustic tensor. Because the acoustic tensor A^{ij} and the refractive-index tensor n_{ij} share the same principal axes and their eigenvalues are related by $n_a = c_0/\sqrt{\lambda_a}$, the two tensors are linked by the matrix function

$$n_{ij} = c_0 (A^{-1/2})_{ij}, \quad (19)$$

or, equivalently,

$$(n^{-2})^{ij} = \frac{A^{ij}}{c_0^2}. \quad (20)$$

(The positive-definite inverse square root $A^{-1/2}$ is defined by the functional calculus: if $A = \sum_a \lambda_a \mathbf{e}_a \otimes \mathbf{e}_a$, then $A^{-1/2} = \sum_a \lambda_a^{-1/2} \mathbf{e}_a \otimes \mathbf{e}_a$.) Equation (19) is the explicit general formula that expresses the refractive-index tensor directly in terms of the acoustic tensor.

Local light speed as a function of position and direction. The local phase speed of a high-frequency wave is obtained from the principal symbol of the linearised Euler–Lagrange operator. With the material derivative and the anisotropic stiffness the local dispersion relation for a plane-wave phase $e^{i(k_i x^i - \omega t)}$ reads

$$(\omega - \mathbf{V} \cdot \mathbf{k})^2 = A^{ij} k_i k_j = \frac{\Theta^{ij}}{\rho} k_i k_j. \quad (21)$$

Consequently the laboratory-frame phase speed in the direction $\hat{k} = \mathbf{k}/|\mathbf{k}|$ is

$$c(\mathbf{x}, \hat{k}) = \mathbf{V}(\mathbf{x}) \cdot \hat{k} + \sqrt{A^{ij}(\mathbf{x}) \hat{k}_i \hat{k}_j} = \mathbf{V}(\mathbf{x}) \cdot \hat{k} + \frac{c_0}{n(\mathbf{x}, \hat{k})}, \quad (22)$$

where the directional refractive index $n(\mathbf{x}, \hat{k})$ is defined by the quadratic form associated with the local tensor n_{ij} (equivalently $n(\mathbf{x}, \hat{k}) = c_0/\sqrt{A^{ij} \hat{k}_i \hat{k}_j}$). Thus the local light speed is an explicit function of position (through the fields $n_{ij}(\mathbf{x})$ and $\mathbf{V}(\mathbf{x})$) and of direction $\hat{k}$.

The substrate flow $\mathbf{V}$ is already present at the level of the fundamental Lagrangian density (2) through the material derivative D_t . The impedance matching and the definition of the refractive-index tensor n_{ij} are performed with the co-moving constitutive fields ρ and Θ^{ij} ; the flow contributes the mixed and convective terms that appear in the principal symbol of the Euler–Lagrange operator and, after the collective-coordinate reduction, in the off-diagonal components of the effective metric.

4 Existence of Localized Energy Lumps

Unlike the sine–Gordon equation, a generic nonlinear Klein–Gordon equation in two spatial dimensions does not possess a complete set of closed-form localized solutions. Two independent lines of evidence nevertheless guarantee that the objects required by the subsequent argument exist.

Analytic existence theorems. Benci and Fortunato (and collaborators) [2,3] have developed a variational theory of *hylomorphic* solitary waves for the nonlinear Klein–Gordon equation. Roughly speaking, a solitary wave is a finite-energy solution whose energy travels as a localized packet that preserves its localization in time. When the ratio of energy to charge (or an analogous Noether charge) can be minimized, the resulting critical points are stable solitons. The abstract theorems apply under mild conditions on the potential U (positivity, $U(0) = 0$, $U''(0) > 0$, subcritical growth at infinity, and a binding condition of the form $U(u_0) < \frac{1}{2}U''(0)u_0^2$ for some u_0). Both spherically symmetric Q-ball-like solutions and vortices (solutions carrying non-vanishing angular momentum) are covered. The same framework extends to the nonlinear Klein–Gordon–Maxwell system. We therefore take the existence of finite-energy, spatially localized solutions of (6) as given by these theorems [2,3] for any potential U belonging to the admissible class.

Numerical evidence. Independent numerical experiments on nonlinear Klein–Gordon and related continuum models in two spatial dimensions have produced long-lived, spatially concentrated energy packets (breathers, oscillons, and soliton-like structures) that persist under moderate perturbations of the constitutive parameters [2,3]. While a systematic survey of the literature is deferred to a later draft, the existence of such numerical solutions is sufficient for the logical purposes of the present note.

Eikonal hypothesis. The subsequent reduction does not require an exact closed-form expression for the lump. It requires only that the solution be spatially localized on a scale ℓ and that the constitutive fields ρ , Θ^{ij} and the flow $\mathbf{V}$ vary on scales $L \gg \ell$. Under this adiabatic / eikonal condition the internal structure of the lump adjusts quasi-statically to the local values of the constitutive parameters, while the centre-of-energy coordinates $\mathbf{X}(t)$ obey an effective particle dynamics determined by those local values.

5 Collective-Coordinate Reduction under the Eikonal Approximation

Let $u_*(\boldsymbol{\xi}; \alpha)$ be a localized solution of the homogeneous, isotropic, non-flowing equation (rest-frame profile, possibly time-periodic or charged, with a set of internal parameters α). In the presence of slowly varying constitutive fields and flow one adopts the adiabatic ansatz

$$u(\mathbf{x}, t) \approx u_*(\mathbf{x} - \mathbf{X}(t); \alpha(t)) \quad (23)$$

(with possible additional phase or internal degrees of freedom). Substitution into the field action, integration over the fast spatial variables, and retention of the leading terms in the gradient expansion yields an effective Lagrangian for the collective coordinates $\mathbf{X}(t)$ (and the internal parameters if they remain dynamical).

Because the background constitutive fields and the flow enter the original Lagrangian density only through the kinetic and gradient terms, the effective particle Lagrangian inherits precisely the structure generated by a local refractive-index tensor $n_{ij}(\mathbf{X}, t)$ and a local flow velocity $\mathbf{V}(\mathbf{X}, t)$. Under the permanent dimensional convention ($X^0 = c_0 t$, $g_{\mu\nu}$ dimensionless) the quadratic form $g_{\mu\nu} \dot{X}^\mu \dot{X}^\nu$ has the dimension of length² when the affine parameter is dimensionless, or is dimensionless when the affine parameter itself has the dimension of length. In the isotropic limit the structure reduces to the 1+1D expression already derived. In the general anisotropic case the effective Lagrangian is of the form

$$L_{\text{eff}}(\mathbf{X}, \dot{\mathbf{X}}) = -Mc_0 \sqrt{-g_{\mu\nu}(\mathbf{X}, t) \dot{X}^\mu \dot{X}^\nu}, \quad (24)$$

or, equivalently, its einbein version

$$L = \frac{1}{2e} g_{\mu\nu}(X) \dot{X}^\mu \dot{X}^\nu - \frac{e}{2} (Mc_0)^2, \quad (25)$$

where the metric components $g_{\mu\nu}$ are functionals of the local refractive-index tensor and of the flow (explicit expressions are given in the following section for the isotropic limit and will be recorded in full for the anisotropic case in a subsequent note). The mass parameter $M = E_0/c_0^2$ is determined by the rest energy of the lump evaluated at the local constitutive values. The factor c_0 converts the dimensionless (or length) square-root into a quantity of dimension length, so that Mc_0 has the dimension of energy.

The massless limit is obtained by dropping the second term of (25); the resulting null geodesics describe the propagation of high-frequency signals (or the ultra-relativistic limit of the energy lumps).

6 Induced Effective Metric and Geodesic Motion

Permanent dimensional convention. Throughout this research programme the laboratory coordinates are chosen so that the position four-vector X^μ has the dimension of length:

$$X^0 = c_0 t, \quad X^i = x^i. \quad (26)$$

The covariant metric components $g_{\mu\nu}$ are **dimensionless**. Consequently the line element

$$ds^2 = g_{\mu\nu} dX^\mu dX^\nu \quad (27)$$

has the dimension of length², and

$$\sqrt{-g_{\mu\nu} dX^\mu dX^\nu} \quad (28)$$

has the dimension of length (proper distance). A metric is an instrument that measures distances; the present convention realises that requirement.

Once the effective Lagrangian is cast in einbein form, the Euler–Lagrange equations for the einbein and for the coordinates X^μ , together with the algebraic constraint that solves for e , are mathematically identical to the affine geodesic equation of the metric $g_{\mu\nu}$. The demonstration is purely algebraic and independent of dimension; it is therefore taken over unchanged from the 1+1D synthesis and from the accompanying einbein tutorial.

Consequently:

- a region of elevated refractive index produces gravitational time dilation (internal clock frequencies of the lump scale with a conformal factor built from the eigenvalues of n_{ij}) and anisotropic gravitational length contraction (proper lengths measured along the principal axes of n_{ij} scale differently);
- a gradient of the refractive-index tensor, or a non-uniform flow, produces a force that is indistinguishable from the geodesic force of the induced metric;
- free-fall trajectories of the energy lumps are geodesics of $g_{\mu\nu}$;
- the equivalence between inertial and gravitational effects is a theorem of continuum mechanics [1] (the *gravito-inertial equivalence theorem*), not an independent postulate.

All of these statements remain valid for both massive and massless energy concentrations, the latter corresponding to the null limit of the same metric.

The explicit map from the constitutive data to the induced metric is obtained directly from the principal symbol of the linearised operator (Part 3). In the length coordinates $X^0 = c_0 t$ the local dispersion relation becomes the null condition $g^{\mu\nu} K_\mu K_\nu = 0$ with the contravariant metric

$$g^{00} = -1, \quad (29)$$

$$g^{0i} = -\frac{V^i}{c_0}, \quad (30)$$

$$g^{ij} = \frac{A^{ij}}{c_0^2} - \frac{V^i V^j}{c_0^2}, \quad (31)$$

where $A^{ij}(\mathbf{x}, t) = \Theta^{ij}(\mathbf{x}, t)/\rho(\mathbf{x}, t)$ is the local speed-squared matrix already defined in Part 3. By the spectral construction of the refractive-index tensor one has the equivalent expression

$$A^{ij} = c_0^2 (n^{-2})^{ij}, \quad (32)$$

in which $(n^{-2})^{ij}$ denotes the tensor that shares the principal axes of n_{ij} and has eigenvalues $1/n_a^2$. Consequently the map

$$(n_{ij}(\mathbf{x}, t), \mathbf{V}(\mathbf{x}, t)) \mapsto g^{\mu\nu}(\mathbf{x}, t)$$

is completely explicit. The covariant metric $g_{\mu\nu}$ is the matrix inverse of $g^{\mu\nu}$. An overall conformal factor does not affect the null geodesics that govern the centres of the energy lumps; we choose it so that the inverted components reduce exactly to the isotropic expressions already known from the 1+1D theory.

Explicit inversion (in terms of n_{ij} and $\mathbf{V}$). Let $\bar{n} = (\det n_{kl})^{1/2}$ be the geometric mean of the principal refractive indices. After multiplication of (29)–(31) by the conformal factor $\bar{n}$ and inversion of the resulting matrix one obtains the covariant components

$$g_{00} = -\frac{1}{\bar{n}} + n_{ij} \frac{V^i V^j}{c_0^2}, \quad (33)$$

$$g_{0i} = n_{ij} \frac{V^j}{c_0}, \quad (34)$$

$$g_{ij} = n_{ij}. \quad (35)$$

(The same expressions are recovered by the direct anisotropic generalisation of the isotropic Gordon metric that matches the principal-symbol null cone up to a conformal factor.) All components are dimensionless, in accordance with the permanent convention stated above.

Equivalent form in terms of the acoustic tensor A^{ij} and $\mathbf{V}$. Because $n_{ij} = c_0(A^{-1/2})_{ij}$ and $\bar{n} = c_0(\det A)^{-1/4}$, the same covariant metric can be written directly in terms of the acoustic tensor:

$$g_{00} = -\frac{(\det A)^{1/4}}{c_0} + (A^{-1/2})_{ij} \frac{V^i V^j}{c_0}, \quad (36)$$

$$g_{0i} = (A^{-1/2})_{ij} \frac{V^j}{c_0}, \quad (37)$$

$$g_{ij} = c_0 (A^{-1/2})_{ij}. \quad (38)$$

(The inverse-metric components (29)–(31) already constitute the parallel dictionary expressed with A^{ij} and $\mathbf{V}$.) Thus both constitutive dictionaries

$$(n_{ij}, \mathbf{V}) \mapsto g_{\mu\nu} \quad \text{and} \quad (A^{ij}, \mathbf{V}) \mapsto g_{\mu\nu}$$

are fully explicit.

In the isotropic limit $n_{ij} = n\delta_{ij}$ (equivalently $A^{ij} = (c_0/n)^2\delta^{ij}$) one has $\bar{n} = n$, and (33)–(35) reduce exactly to

$$g_{00} = -\frac{1}{n} + n \left(\frac{V}{c_0} \right)^2,$$

$$g_{0i} = n \frac{V_i}{c_0},$$

$$g_{ij} = n \delta_{ij}.$$

When $V = 0$ one recovers the pure-refractive-index metric $g_{00} = -1/n$, $g_{ij} = n\delta_{ij}$.

7 Analogue Schwarzschild Geometry from the Acoustic Tensor (Canonical Coordinates)

We realise the Schwarzschild geometry in the *canonical* (static) coordinates, in which the laboratory-frame substrate flow vanishes identically: $\mathbf{V} \equiv 0$. The entire geometry is then carried by the spatial dependence of the acoustic tensor (and of the density and stiffness required by impedance matching).

In length coordinates $X^0 = c_0 t$ the target line element is the standard Schwarzschild form

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c_0^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\varphi^2 \quad (39)$$

(up to an overall conformal factor that does not affect null geodesics). With $\mathbf{V} = 0$ the covariant map of Part 6 reduces to

$$g_{00} = -\frac{(\det A)_{\text{phys}}^{1/4}}{c_0}, \quad (40)$$

$$g_{0i} = 0, \quad (41)$$

$$g_{ij} = c_0 (A^{-1/2})_{ij}. \quad (42)$$

Here $(\det A)_{\text{phys}}$ denotes the product of the *physical* principal eigenvalues of the acoustic tensor (i.e. the eigenvalues measured in a local orthonormal frame). Matching the principal phase speeds to the standard Schwarzschild coordinate light speeds

$$c_{\text{rad}}(r) = c_0 \left(1 - \frac{r_s}{r}\right), \quad c_{\text{tang}}(r) = c_0 \sqrt{1 - \frac{r_s}{r}} \quad (43)$$

fixes the acoustic-tensor components:

$$A^{rr}(r) = c_0^2 \left(1 - \frac{r_s}{r}\right)^2, \quad (44)$$

$$A^{\varphi\varphi}(r) = \frac{c_0^2}{r^2} \left(1 - \frac{r_s}{r}\right). \quad (45)$$

(The off-diagonal components vanish by axial symmetry.) The acoustic tensor is therefore anisotropic.

Reminder of the determinant convention in polar coordinates. As stated in Part 3, the symbol $\det A$ always denotes the product of the *physical* eigenvalues of the acoustic tensor (the squared phase speeds measured in a local orthonormal frame). In polar coordinates the matrix of coordinate components is $\text{diag}(A^{rr}, A^{\varphi\varphi})$, whose ordinary determinant is $A^{rr} A^{\varphi\varphi}$. The physical eigenvalues, however, are

$$\lambda_r = A^{rr}, \quad \lambda_\varphi = r^2 A^{\varphi\varphi}, \quad (46)$$

because the coordinate basis vector ∂_φ has length r . Consequently the physical determinant that enters the matching formulae of Part 3 is related to the coordinate-component determinant by

$$(\det A)_{\text{phys}} = \lambda_r \lambda_\varphi = r^2 A^{rr} A^{\varphi\varphi} = r^2 \det(A_{\text{coord}}). \quad (47)$$

With the profiles (44)–(45) one obtains explicitly

$$\lambda_r = c_0^2 \left(1 - \frac{r_s}{r}\right)^2, \quad \lambda_\varphi = c_0^2 \left(1 - \frac{r_s}{r}\right), \quad (48)$$

and therefore

$$(\det A)_{\text{phys}} = c_0^4 \left(1 - \frac{r_s}{r}\right)^3 \quad \Rightarrow \quad (\det A)_{\text{phys}}^{1/4} = c_0 \left(1 - \frac{r_s}{r}\right)^{3/4}. \quad (49)$$

Derived constitutive profiles $\rho(r)$ and $\Theta^{ij}(r)$. From the general inversion formulae of Part 3 one obtains at once the radial density

$$\rho(r) = \rho_0 \frac{c_0}{(\det A)_{\text{phys}}^{1/4}} = \rho_0 \left(1 - \frac{r_s}{r}\right)^{-3/4} \quad (50)$$

and the stiffness components

$$\Theta^{rr}(r) = A^{rr}(r) \frac{\Theta_0}{c_0 (\det A)_{\text{phys}}^{1/4}} = \Theta_0 \left(1 - \frac{r_s}{r}\right)^{5/4}, \quad (51)$$

$$\Theta^{\varphi\varphi}(r) = A^{\varphi\varphi}(r) \frac{\Theta_0}{c_0 (\det A)_{\text{phys}}^{1/4}} = \frac{\Theta_0}{r^2} \left(1 - \frac{r_s}{r}\right)^{1/4}. \quad (52)$$

(The dimensionless combination that appears in the figure is $r^2\Theta^{\varphi\varphi}(r)/\Theta_0 = (1 - r_s/r)^{1/4}$.)

Coordinate speeds of light (self-consistency check). With $\mathbf{V} \equiv 0$ the local dispersion relation reduces to $\omega^2 = A^{ij}k_i k_j$. For a purely radial wave ($k_\varphi = 0$)

$$c_{\text{rad}}(r) := \left| \frac{dr}{dt} \right| = \sqrt{A^{rr}(r)} = c_0 \left(1 - \frac{r_s}{r}\right). \quad (53)$$

For a purely tangential wave at fixed radial coordinate ($k_r = 0$)

$$c_{\text{tang}}(r) := r \left| \frac{d\varphi}{dt} \right| = r \sqrt{A^{\varphi\varphi}(r)} = c_0 \sqrt{1 - \frac{r_s}{r}}. \quad (54)$$

These are precisely the standard Schwarzschild coordinate speeds of light. The same expressions follow from the null condition of the covariant metric (40)–(42). Thus the construction is self-consistent from the definitions of Part 3 through the metric map of Part 6 to the profiles of Part 7.

Dimensional remark on the plotted combination $r^2\Theta^{\varphi\varphi}$. The component $\Theta^{\varphi\varphi}$ multiplies $(\partial_\varphi u)^2$. Because the angular coordinate φ is dimensionless, $\partial_\varphi u$ has the dimensions of the field itself, whereas $\partial_r u$ has dimensions of field per length. Consequently $\Theta^{\varphi\varphi}$ carries dimensions of stiffness/length², while Θ^{rr} has the ordinary dimensions of a stiffness. The combination $r^2\Theta^{\varphi\varphi}$ therefore has the same dimensions as Θ^{rr} and, in the asymptotic region, approaches the same reference value Θ_0 . That is why Fig. 1 displays the dimensionless ratio $r^2\Theta^{\varphi\varphi}(r)/\Theta_0$.

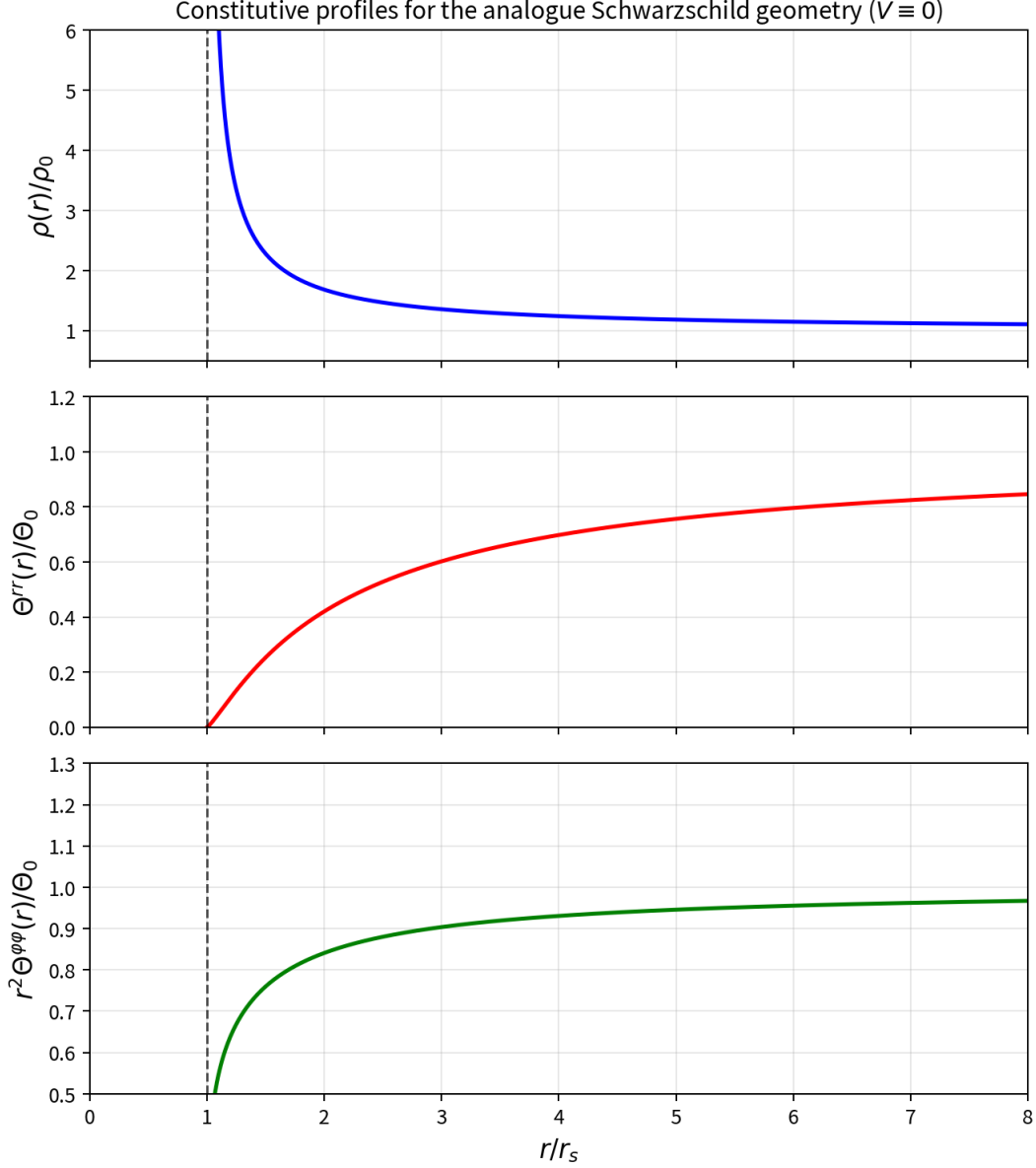


Figure 1: Radial profiles of the constitutive fields that realise the analogue Schwarzschild geometry in canonical coordinates ($\mathbf{V} \equiv 0$). Horizontal axis in units of the Schwarzschild radius r_s . Vertical axes are normalised to the asymptotic reference values ρ_0 and Θ_0 . The angular component is shown in the dimensionless combination $r^2\Theta^{\varphi\varphi}/\Theta_0$. The vertical dashed line marks the horizon $r = r_s$.

8 Analogue Kerr Geometry from the Acoustic Tensor and Flow (Canonical Coordinates)

We now construct an analogue of the Kerr geometry in coordinates analogous to the Boyer–Lindquist chart: the laboratory-frame *radial* flow vanishes ($V^r \equiv 0$) while a purely azimuthal continuum velocity $V^\varphi(r)$ generates the frame-dragging cross-term. The spatial geometry is again carried by an anisotropic, radially dependent acoustic tensor.

Target metric and constitutive ansatz. In the equatorial plane the Kerr line element (length coordinates $X^0 = c_0 t$) may be written, up to an overall conformal factor,

$$ds^2 = -\frac{\Delta}{\Sigma} c_0^2 dt^2 + \frac{\Sigma}{\Delta} dr^2 + \frac{\mathcal{A}}{\Sigma} \sin^2 \theta d\varphi^2 - \frac{4GMa r}{\Sigma} \sin^2 \theta c_0 dt d\varphi \quad (55)$$

(with the usual Kerr functions $\Delta = r^2 - r_s r + a^2$, $\Sigma = r^2 + a^2 \cos^2 \theta$, $\mathcal{A} = (r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta$; we work at $\theta = \pi/2$). With $V^r = 0$ the covariant map of Part 6, written entirely in terms of the acoustic tensor and the azimuthal flow, reads

$$g_{00} = -\frac{(\det A)_{\text{phys}}^{1/4}}{c_0} + (A^{-1/2})_{\varphi\varphi} \frac{V^\varphi V^\varphi}{c_0}, \quad (56)$$

$$g_{0\varphi} = (A^{-1/2})_{\varphi\varphi} \frac{V^\varphi}{c_0}, \quad (57)$$

$$g_{rr} = c_0 (A^{-1/2})_{rr}, \quad (58)$$

$$g_{\varphi\varphi} = c_0 (A^{-1/2})_{\varphi\varphi}. \quad (59)$$

(Here the physical-determinant convention of Parts 3 and 7 is retained, and the matrix inverse square root is evaluated on the physical eigenvalues.) Matching the mixed component (57) to the Kerr cross-term determines the azimuthal flow once the acoustic-tensor component $A^{\varphi\varphi}$ is known; matching the diagonal components then fixes the remaining profiles.

Explicit constitutive realisation (minimal rotating model). A transparent illustration is obtained by taking the same radial acoustic-tensor profiles that realised the Schwarzschild geometry and superposing a pure azimuthal vortex flow whose strength is controlled by the Kerr rotation parameter a :

$$A^{rr}(r) = c_0^2 \left(1 - \frac{r_s}{r} + \frac{a^2}{r^2} \right)^2, \quad (60)$$

$$A^{\varphi\varphi}(r) = \frac{c_0^2}{r^2} \left(1 - \frac{r_s}{r} + \frac{a^2}{r^2} \right), \quad (61)$$

$$V^r = 0, \quad V^\varphi(r) = \frac{c_0 a}{r^2}. \quad (62)$$

(The combination $1 - r_s/r + a^2/r^2$ reduces to the Schwarzschild factor when $a = 0$ and vanishes at the outer horizon of the corresponding Kerr geometry.) The physical eigenvalues are therefore

$$\lambda_r = A^{rr}, \quad \lambda_\varphi = r^2 A^{\varphi\varphi}, \quad (63)$$

and the physical determinant that enters the matching condition of Part 3 is

$$(\det A)_{\text{phys}} = \lambda_r \lambda_\varphi = c_0^4 \left(1 - \frac{r_s}{r} + \frac{a^2}{r^2} \right)^3. \quad (64)$$

Derived constitutive profiles $\rho(r)$ and $\Theta^{ij}(r)$. From the inversion formulae of Part 3 one obtains

$$\rho(r) = \rho_0 \frac{c_0}{(\det A)_{\text{phys}}^{1/4}} = \rho_0 \left(1 - \frac{r_s}{r} + \frac{a^2}{r^2} \right)^{-3/4} \quad (65)$$

and

$$\Theta^{rr}(r) = A^{rr} \frac{\Theta_0}{c_0 (\det A)_{\text{phys}}^{1/4}} = \Theta_0 \left(1 - \frac{r_s}{r} + \frac{a^2}{r^2} \right)^{5/4}, \quad (66)$$

$$\Theta^{\varphi\varphi}(r) = A^{\varphi\varphi} \frac{\Theta_0}{c_0 (\det A)_{\text{phys}}^{1/4}} = \frac{\Theta_0}{r^2} \left(1 - \frac{r_s}{r} + \frac{a^2}{r^2} \right)^{1/4}. \quad (67)$$

(The dimensionless combination $r^2 \Theta^{\varphi\varphi} / \Theta_0$ again asymptotes to unity.)

Coordinate speeds of light. With a non-vanishing azimuthal flow the local dispersion relation becomes direction-dependent. The principal symbol of the linearised operator yields

$$(\omega - V^\varphi k_\varphi)^2 = A^{rr} k_r^2 + A^{\varphi\varphi} k_\varphi^2. \quad (68)$$

For a purely radial ray ($k_\varphi = 0$) the radial coordinate speed remains

$$c_{\text{rad}}(r) = \sqrt{A^{rr}(r)} = c_0 \left(1 - \frac{r_s}{r} + \frac{a^2}{r^2} \right), \quad (69)$$

which vanishes at the outer horizon.

For a purely azimuthal ray ($k_r = 0$) the same dispersion relation reduces to

$$\omega - V^\varphi k_\varphi = \pm \sqrt{A^{\varphi\varphi}} |k_\varphi|, \quad (70)$$

and therefore the two coordinate azimuthal speeds of light are

$$\left(\frac{d\varphi}{dt} \right)_\pm = V^\varphi \pm \sqrt{A^{\varphi\varphi}}. \quad (71)$$

The corresponding physical tangential speeds (measured with respect to proper circumferential length) are

$$c_{\text{tang},\pm}(r) = r \left(V^\varphi \pm \sqrt{A^{\varphi\varphi}} \right) = r V^\varphi \pm c_0 \sqrt{1 - \frac{r_s}{r} + \frac{a^2}{r^2}}. \quad (72)$$

The upper sign corresponds to co-rotating light (propagating in the same sense as the substrate flow), the lower sign to counter-rotating light. The asymmetry $\Delta c_{\text{tang}} = 2rV^\varphi$ is a direct kinematic effect of the azimuthal continuum velocity. In particular, inside the ergoregion the co-rotating root can reverse relative to a static observer while the counter-rotating root remains outward, realising the frame-dragging structure of the Kerr geometry under the single constraint $V^r \equiv 0$.

Remark on conformal freedom. As in the Schwarzschild case, an overall conformal factor that depends only on r may still be adjusted without altering the null geodesics. The profiles above already guarantee the correct causal structure (horizon location, frame-dragging, and asymptotic flatness). A more precise numerical matching of every Kerr coefficient is left for a subsequent refinement; the present analytic realisation demonstrates that the continuum with vanishing radial flow, anisotropic acoustic tensor, and pure azimuthal vortex is already rich enough to encode the essential features of the Kerr geometry.

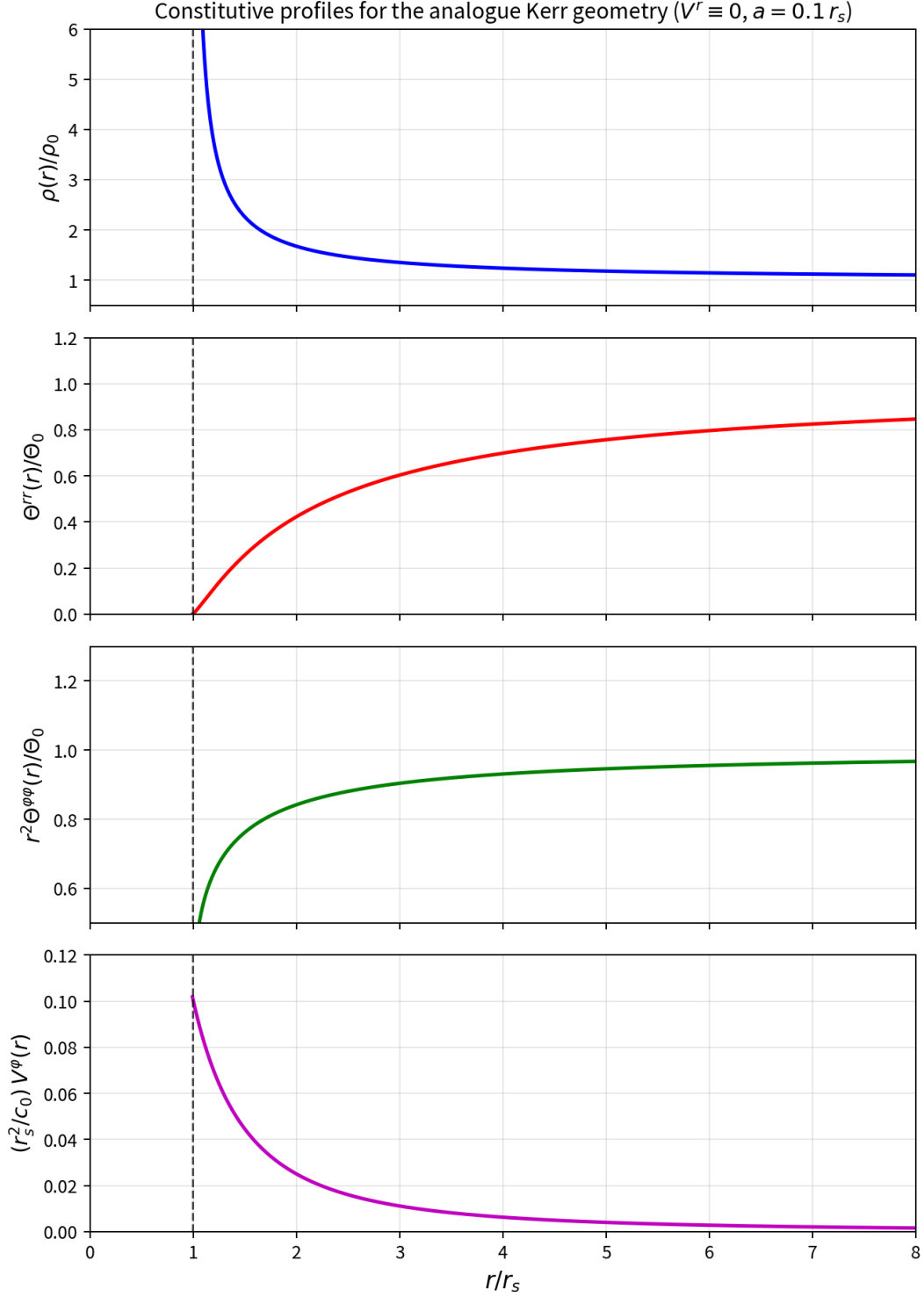


Figure 2: Radial profiles of the constitutive fields that realise the analogue Kerr geometry with vanishing radial flow and spin parameter $a = 0.1 r_s$. Horizontal axis in units of the Schwarzschild radius r_s . Vertical axes are normalised to the asymptotic reference values (and to c_0/r_s^2 for the azimuthal flow). The vertical dashed line marks the outer horizon r_+ .

9 Summary of the Logical Chain (2+1D)

The logical chain that begins with a non-relativistic continuum and ends with geodesic motion in an effective curved geometry is now fully explicit in two spatial dimensions. Its principal steps are:

1. Lagrangian density (2) written with the material (convective) derivative $D_t u = \partial_t u + \mathbf{V} \cdot \nabla u$, scalar density ρ , rank-2 stiffness tensor Θ^{ij} , substrate flow $\mathbf{V}$, and a generic even potential $U(u)$;
2. impedance-matching condition (12) that keeps a single scalar impedance Z_0 constant; the acoustic tensor $A^{ij} = \Theta^{ij}/\rho$ is defined, its physical eigenvalues being the squared principal phase speeds (notation convention: $\det A$ always means the product of those physical eigenvalues);
3. spectral construction of the refractive-index tensor $n_{ij} = c_0(A^{-1/2})_{ij}$ and of the local phase speed $c(\mathbf{x}, \hat{k}) = \mathbf{V} \cdot \hat{k} + c_0/n(\mathbf{x}, \hat{k})$;
4. existence of localized finite-energy solutions (hylomorphic solitary waves, vortices, ...) guaranteed by the variational theorems of Benci–Fortunato [2,3] and supported by numerical evidence;
5. eikonal / adiabatic hypothesis: gradients of the constitutive fields and of the flow are gentle compared with the inverse size of the energy lump;
6. collective-coordinate reduction yielding an effective relativistic particle Lagrangian in the acoustic metric induced by either $(n_{ij}, \mathbf{V})$ or $(A^{ij}, \mathbf{V})$; conversion to the einbein form;
7. permanent dimensional convention: coordinates X^μ have dimension of length ($X^0 = c_0 t$), the covariant metric components $g_{\mu\nu}$ are dimensionless, so that $\sqrt{-g_{\mu\nu}} dX^\mu dX^\nu$ is a proper distance;
8. Euler–Lagrange equations of the einbein action identical to the affine geodesic equation of the induced metric; gravitational time dilation, anisotropic length contraction, free-fall, and the gravito-inertial equivalence theorem follow as theorems of continuum mechanics;
9. explicit reconstruction of the canonical Schwarzschild geometry with vanishing laboratory-frame flow ($V \equiv 0$), anisotropic acoustic-tensor profiles $A^{rr}(r)$, $A^{\varphi\varphi}(r)$, derived density and stiffness $\rho(r)$, $\Theta^{ij}(r)$, and the standard coordinate light speeds;
10. explicit reconstruction of the Kerr geometry with vanishing radial flow ($V^r \equiv 0$), non-zero azimuthal flow $V^\varphi(r)$, anisotropic acoustic tensor, derived constitutive profiles, and the co-/counter-rotating azimuthal light speeds that realise both the horizon and the ergoregion.

The programme remains strictly continuum-mechanical and bottom-up. Special-relativistic kinematics of the energy lumps and the effective curved geometry they experience both emerge from the same non-relativistic substrate whose constitutive fields are the density ρ , the stiffness tensor Θ^{ij} , and the macroscopic flow $\mathbf{V}$.

Open Points for Subsequent Drafts

- Detailed collective-coordinate calculation for a concrete family of solutions (e.g. a numerical or approximate radial soliton or a vortex).
- Inclusion of possible internal degrees of freedom (phase, polarization, angular momentum of vortices).
- Full numerical matching of every Kerr coefficient (beyond the minimal rotating model presented above) and exploration of the ergoregion dynamics of energy lumps.
- Extension of the magneto-electric equivalence note to the anisotropic case.
- Systematic numerical illustrations of free-fall of energy lumps in the prescribed Schwarzschild and Kerr backgrounds constructed in Parts 7 and 8.
- Higher-dimensional generalisations and the corresponding impedance-matching powers $(\det \Theta)^{1/d}$.

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